EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Four points $A(+1, -1, 1)$; $B(1, 3, 1)$; $C(4, 3, 1)$ and $D(4, -1, 1)$ taken in order are the vertices of
(A) a parallelogram which is neither a rectangle nor a rhombus
(B) rhombus
(C) an isosceles trapezium
(D) a cyclic quadrilateral

MVT097

2. Let α, β & γ be distinct real numbers. The points whose position vector's are $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$; $\beta\hat{i} + \gamma\hat{j} + \alpha\hat{k}$ and $\gamma\hat{i} + \alpha\hat{j} + \beta\hat{k}$

(A) are collinear
(B) form an equilateral triangle
(C) form a scalene triangle
(D) form a right angled triangle

MVT098

3. Let $A(0, -1, 1)$, $B(0, 0, 1)$, $C(1, 0, 1)$ are the vertices of a ΔABC. If R and r denotes the circumradius and inradius of ΔABC, then $\frac{r}{R}$ has value equal to

(A) $\tan\frac{3\pi}{8}$ (B) $\cot\frac{3\pi}{8}$ (C) $\tan\frac{\pi}{12}$ (D) $\cot\frac{\pi}{12}$

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MVT099

4. Consider the points A, B and C with position vectors $\left(-2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ and $7\hat{i}-\hat{k}$ respectively.

Statement-1: The vector sum, $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA} = \vec{0}$

because

Statement-2: A, B and C form the vertices of a triangle.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

MVT100

5. If the vector $6\hat{i} - 3\hat{j} - 6\hat{k}$ is decomposed into vectors parallel and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ then the vectors are :

(A) $-\left(\hat{i} + \hat{j} + \hat{k}\right)$ & $7\hat{i} - 2\hat{j} - 5\hat{k}$
(B) $-2\left(\hat{i} + \hat{j} + \hat{k}\right)$ & $8\hat{i} - \hat{j} - 4\hat{k}$
(C) $+2\left(\hat{i} + \hat{j} + \hat{k}\right)$ & $4\hat{i} - 5\hat{j} - 8\hat{k}$
(D) none

MVT101

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6. A, B, C & D are four points in a plane with pv's $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ respectively such that $(\vec{a}-\vec{d})\cdot(\vec{b}-\vec{c})=(\vec{b}-\vec{d})\cdot(\vec{c}-\vec{a})$ = 0. Then for the triangle ABC, D is its

(A) incentre (B) circumcentre (C) orthocentre (D) centroid

MVT102

7. Let $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{a}+\hat{b}+\hat{c}$ is also a unit vector. If pairwise angles between $\hat{a}, \hat{b}, \hat{c}$ are θ_1, θ_2 and θ_3 respectively then $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$ equals

(A) 3 (B) – 3 (C) 1 (D) – 1

MVT103

8. A tangent is drawn to the curve $y = \frac{8}{x^2}$ at a point $A(x_1, y_1)$, where $x_1 = 2$. The tangent cuts the x-axis at point B. Then the scalar product of the vectors $\vec{AB}$ & $\vec{OB}$ is

(A) 3 (B) – 3 (C) 6 (D) – 6

MVT104

9. Cosine of an angle between the vectors $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ if $|\vec{a}| = 2$, $|\vec{b}| = 1$ and $\vec{a}\wedge\vec{b} = 60°$ is

(A) $\sqrt{3/7}$ (B) $9/\sqrt{21}$ (C) $3/\sqrt{7}$ (D) none

MVT105

10. The vector equations of two lines L_1 and L_2 are respectively

$\vec{r}=17\hat{i}-9\hat{j}+9\hat{k}+\lambda(3\hat{i}+\hat{j}+5\hat{k})$ and $\vec{r}=15\hat{i}-8\hat{j}-\hat{k}+\mu(4\hat{i}+3\hat{j})$

I L_1 and L_2 are skew lines

II (11, –11, –1) is the point of intersection of L_1 and L_2

III (–11, 11, 1) is the point of intersection of L_1 and L_2

IV $\cos^{-1}\left(3/\sqrt{35}\right)$ is the acute angle between L_1 and L_2

then , which of the following is true?

(A) II and IV (B) I and IV (C) IV only (D) III and IV

MVT106

11. Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}|=1, |\vec{v}|=2, |\vec{w}|=3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and vectors $\vec{v}$, $\vec{w}$ are perpendicular to each other then $|\vec{u}-\vec{v}+\vec{w}|$ equals

(A) 2 (B) $\sqrt{7}$ (C) $\sqrt{14}$ (D) 14

MVT107

12. Given an equilateral triangle ABC with side length equal to 'a'. Let M and N be two points respectively on the side AB and AC such that $\overrightarrow{AN} = K\overrightarrow{AC}$ and $\overrightarrow{AM} = \frac{\overrightarrow{AB}}{3}$. If $\overrightarrow{BN}$ and $\overrightarrow{CM}$ are orthogonal then the value of K is equal to

(A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

MVT108

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13. If $\vec{p}$ & $\vec{s}$ are not perpendicular to each other and $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ & $\vec{r} \cdot \vec{s} = 0$, then $\vec{r}$ =

(A) $\vec{p} \cdot \vec{s}$ (B) $\vec{q} + \left(\frac{\vec{q} \cdot \vec{p}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$ (C) $\vec{q} - \left(\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$ (D) $\vec{q} + \mu\vec{p}$ for all scalars μ

MVT109

14. Given a parallelogram $OACB$. The lengths of the vectors $\vec{OA}$, $\vec{OB}$ & $\vec{AB}$ are a, b & c respectively. The scalar product of the vectors $\vec{OC}$ & $\vec{OB}$ is :

(A) $\frac{a^2 - 3b^2 + c^2}{2}$ (B) $\frac{3a^2 + b^2 - c^2}{2}$ (C) $\frac{3a^2 - b^2 + c^2}{2}$ (D) $\frac{a^2 + 3b^2 - c^2}{2}$

MVT110

15. Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \frac{2\pi}{3}$. If $|\vec{a}| = 1$, $|\vec{b}| = 2$ then $\left\{\left(\vec{a} + 3\vec{b}\right) \times \left(3\vec{a} - \vec{b}\right)\right\}^2$ =

(A) 225 (B) 250 (C) 275 (D) 300

MVT111

16. The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ & $\vec{c} = 3\hat{i} + \hat{j} + 4\hat{k}$ are so placed that the end point of one vector is the starting point of the next vector. Then the vectors are -

(A) not coplanar

(B) coplanar but cannot form a triangle

(C) coplanar but can form a triangle

(D) coplanar & can form a right angled triangle

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MVT112

17. Given the vectors

$\vec{u} = 2\hat{i} - \hat{j} - \hat{k}$

$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$

$\vec{w} = \hat{i} - \hat{k}$

If the volume of the parallelopiped having $-c\vec{u}$, $\vec{v}$ and $c\vec{w}$ as concurrent edges, is 8 then 'c' can be equal to

(A) ± 2 (B) 4 (C) 8 (D) can not be determined

MVT113

18. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ & $\vec{b}$. If the angle between $\vec{a}$ & $\vec{b}$ is $\frac{\pi}{6}$, then

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 =$$

(A) 0 (B) 1

(C) $(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$ (D) $(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2)$

MVT114

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19. Given 3 vectors $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$; $\vec{V}_2 = b\hat{i} + c\hat{j} + a\hat{k}$; $\vec{V}_3 = c\hat{i} + a\hat{j} + b\hat{k}$

In which one of the following conditions $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ are linearly independent?

(A) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(D) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

MVT115

20. The altitude of a parallelopiped whose three coterminous edges are the vectors, $\vec{A} = \hat{i} + \hat{j} + \hat{k}$; $\vec{B} = 2\hat{i} + 4\hat{j} - \hat{k}$ & $\vec{C} = \hat{i} + \hat{j} + 3\hat{k}$ with $\vec{A}$ and $\vec{B}$ as the sides of the base of the parallelopiped, is

(A) $2/\sqrt{19}$ (B) $4/\sqrt{19}$ (C) $2\sqrt{38}/19$ (D) none

MVT116

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a}\times\vec{b}=\vec{c}$ & $\vec{b}\times\vec{c}=\vec{a}$ then which of the following always hold(s) good?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\left|\vec{b}\right|$

(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\left|\vec{c}\right|^2$ (D) $\left|\vec{b}\right|=\left|\vec{c}\right|$

MVT117

2. If $\vec{A},\vec{B},\vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

(A) $(\vec{A}\times\vec{B})\cdot(\vec{C}\times\vec{D})=0$ (B) $(\vec{A}\times\vec{C})\cdot(\vec{B}\times\vec{D})\neq 0$

(C) $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D})=\vec{0}$ (D) $(\vec{A}\times\vec{C})\times(\vec{B}\times\vec{D})\neq\vec{0}$

MVT118

3. If $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are the pv's of the points A, B, C & D respectively in three dimensional space & satisfy the relation $3\vec{a}-2\vec{b}+\vec{c}-2\vec{d}=\vec{0}$, then :

(A) A, B, C & D are coplanar

(B) the line joining the points B & D divides the line joining the point A & C in the ratio 2 : 1.

(C) the line joining the points A & C divides the line joining the points B & D in the ratio 1 : 1

(D) the four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are linearly dependent.

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MVT119

4. Which of the following statement(s) hold good ?

(A) if $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}\ \Rightarrow\vec{b}=\vec{c}\ \ (\vec{a}\neq\vec{0})$

(B) if $\vec{a}\times\vec{b}=\vec{a}\times\vec{c}\ \Rightarrow\vec{b}=\vec{c}\ \ (\vec{a}\neq\vec{0})$

(C) if $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}$ and $\vec{a}\times\vec{b}=\vec{a}\times\vec{c}\ \Rightarrow\vec{b}=\vec{c}\quad(\vec{a}\neq 0)$

(D) if $\vec{v}_1,\vec{v}_2,\vec{v}_3$ are non coplanar vectors and $\vec{k}_1=\dfrac{\vec{v}_2\times\vec{v}_3}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$; $\vec{k}_2=\dfrac{\vec{v}_3\times\vec{v}_1}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$

and $\vec{k}_3=\dfrac{\vec{v}_1\times\vec{v}_2}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$ then $\vec{k}_1\cdot(\vec{k}_2\times\vec{k}_3)=\dfrac{1}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$

MVT120

5. If a, b, c are different real numbers and $a\hat{i}+b\hat{j}+c\hat{k}$; $b\hat{i}+c\hat{j}+a\hat{k}$ & $c\hat{i}+a\hat{j}+b\hat{k}$ are position vectors of three non-collinear points A, B & C then :

(A) centroid of triangle ABC is $\dfrac{a+b+c}{3}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(B) $\hat{i}+\hat{j}+\hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle.

MVT121

For More Join: @IITJEE_Advanced

6. Let OAB be a regular triangle with side length unity (O being the origin). Also M, N are the points of trisection of AB, M being closer to A and N closer to B. Position vectors of A, B, M and N are $\vec{a}, \vec{b}, \vec{m}$ and $\vec{n}$ respectively. Which of the following hold(s) good ?

(A) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow \frac{2}{3}$ and $y = \frac{1}{3}$

(B) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{5}{6}$ and $y = \frac{1}{6}$

(C) $\vec{m}.\vec{n}$ equals $\frac{13}{18}$

(D) $\vec{m}.\vec{n}$ equals $\frac{15}{18}$

MVT122

7. Given three vectors $\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}$; $\vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$

Which of the following hold good for the vectors $\vec{U}, \vec{V}$ and $\vec{W}$?

(A) $\vec{U}, \vec{V}$ and $\vec{W}$ are linearly dependent

(B) $(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$

(C) $\vec{U}, \vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors

(D) $\vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$

MVT123

8. Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct ?

(A) $\vec{a} \times \vec{b} = [\vec{a}\ \vec{b}\ \hat{i}]\hat{i} + [\vec{a}\ \vec{b}\ \hat{j}]\hat{j} + [\vec{a}\ \vec{b}\ \vec{k}]\hat{k}$

(B) $\vec{a}.\vec{b} = (\vec{a}.\hat{i})(\vec{b}.\hat{i}) + (\vec{a}.\hat{j}).(\vec{b}.\hat{j}) + (\vec{a}.\hat{k})(\vec{b}.\hat{k})$

(C) if $\vec{u} = \hat{a} - (\hat{a}.\hat{b})\hat{b}$ and $\vec{v} = \hat{a} \times \hat{b}$ then $|\vec{u}| = |\vec{v}|$

(D) if $\vec{c} = \vec{a} \times (\vec{a} \times \vec{b})$ and $\vec{d} = \vec{b} \times (\vec{a} \times \vec{b})$ then $\vec{c} + \vec{d} = \vec{0}$

MVT124

9. The value(s) of $\alpha \in [0, 2\pi]$ for which vector $\vec{a} = \hat{i} + 3\hat{j} + (\sin 2\alpha)\hat{k}$ makes an obtuse angle with the z-axis and the vectors $\vec{b} = (\tan\alpha)\hat{i} - \hat{j} + 2\sqrt{\sin\frac{\alpha}{2}}\,\hat{k}$ and $\vec{c} = (\tan\alpha)\hat{i} + (\tan\alpha)\hat{j} - 3\sqrt{\operatorname{cosec}\frac{\alpha}{2}}\,\hat{k}$ are orthogonal, is/are

(A) $\tan^{-1} 3$ (B) $\pi - \tan^{-1} 2$ (C) $\pi + \tan^{-1} 3$ (D) $2\pi - \tan^{-1} 2$

MVT125

10. A line passes through a point A with position vector $3\hat{i} + \hat{j} - \hat{k}$ and is parallel to the vector $2\hat{i} - \hat{j} + 2\hat{k}$. If P is a point on this line such that $AP = 15$ units, then the position vector of the point P is/are

(A) $13\hat{i} + 4\hat{j} - 9\hat{k}$ (B) $13\hat{i} - 4\hat{j} + 9\hat{k}$ (C) $7\hat{i} - 6\hat{j} + 11\hat{k}$ (D) $-7\hat{i} + 6\hat{j} - 11\hat{k}$

MVT126

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11. Which of the followings is/are correct :

(A) The angle between the two straight lines $\vec{r}=3\hat{i}-2\hat{j}+4\hat{k}+\lambda\left(-2\hat{i}+\hat{j}+2\hat{k}\right)$ and $\vec{r}=\hat{i}+3\hat{j}-2\hat{k}+\mu\left(3\hat{i}-2\hat{j}+6\hat{k}\right)$ is $\cos^{-1}\left(\frac{4}{21}\right)$

(B) $(\vec{r}.\hat{i})(\hat{i}\times\vec{r})+(\vec{r}.\hat{j})(\hat{j}\times\vec{r})+(\vec{r}.\hat{k})(\hat{k}\times\vec{r})=\vec{0}$

(C) The force determined by the vector $\vec{r}$ = (1, –8, –7) is resolved along three mutually perpendicular directions, one of which is in the direction of the vector $\vec{a}=2\hat{i}+2\hat{j}+\hat{k}$. Then the vector component of the force $\vec{r}$ in the direction of the vector $\vec{a}$ is $-\frac{7}{3}\left(2\hat{i}+2\hat{j}+\hat{k}\right)$

(D) The cosine of the angle between any two diagonals of a cube is $\frac{1}{3}$.

MVT127

12. $\hat{a}$ and $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$, $\hat{b}$ and $\hat{a}\times\hat{b}$ will be :

(A) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$ (B) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}+\hat{a}\times\hat{b}\right)$

(C) $\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$ (D) $-\frac{1}{\sqrt{3}}\left(\hat{a}+\hat{b}-\hat{a}\times\hat{b}\right)$

MVT128

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13. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d})=\lambda$ and $\vec{a}.\vec{c}=\frac{\sqrt{3}}{2}$, then

(A) $\vec{a},\vec{b},\vec{c}$ are coplanar if $\lambda = 1$

(B) Angle between $\vec{b}$ and $\vec{d}$ is 30° if $\lambda = -1$

(C) angle between $\vec{b}$ and $\vec{d}$ is 150° if $\lambda = -1$

(D) If $\lambda = 1$ then angle between $\vec{b}$ and $\vec{c}$ is 60°

MVT129

14. Let $\bar{p}$, $\bar{q}$, $\bar{r}$ are $\bar{\alpha}=\bar{p}\times(\bar{q}\times\bar{r})$ non zero vectors and $\bar{\beta}=(\bar{p}\times\bar{q})\times\bar{r}$ and if $\bar{\alpha}=\bar{\beta}$ then which of the following holds goods

(A) $\bar{p}$ and $\bar{q}$ are orthogonal. (B) $\bar{p}$ and $\bar{r}$ are collinear

(C) $\bar{q}$ and $\bar{r}$ are orthogonal (D) $\bar{q}=\lambda(\bar{p}\times\bar{r})$ where 'λ' is scalar

MVT130

15. If $|\vec{a}|=|\vec{b}|=|\vec{c}|=|\vec{d}|=1$ such that $(\vec{a}\times\vec{b}).(\vec{c}\times\vec{d})=1$ and $\vec{a}\cdot\vec{c}=\frac{1}{2}$, then which of the following can be correct ?

(A) $\vec{a}\cdot\vec{b}=0$ (B) $\vec{a}\cdot(\vec{b}\times\vec{c})=0$ (C) $\vec{a}\cdot(\vec{b}\times\vec{c})\neq 0$ (D) $|\vec{a}+\vec{b}+\vec{c}|=\sqrt{4+\sqrt{3}}$

MVT131

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question 16 to 17

Vectors are essential tools for understanding and solving problems in physics, engineering and mathematics. One key aspect of working with vectors is finding the angle between them, which can provide valuable information about the relationship between the two vectors. The dot product, also known as the scalar product or inner product, is a crucial operation that allows us to determine the angle between two vectors.

The dot product of two vectors $\vec{A}$ and $\vec{B}$ is defined as :

$\vec{A}\cdot\vec{B}=|\vec{A}||\vec{B}|\cos(\theta)$

where and represent the magnitudes of the vectors A and B, respectively and θ is the angle between the vectors.

From the definition above, we can easily deduce the formula to find the angle between the two vectors :

$\theta=\arccos\left[(\vec{A}\cdot\vec{B})/(|\vec{A}||\vec{B}|)\right]$

16. In a quadrilateral $ABCD$, $\vec{AC}$ is the bisector of the $\left(\vec{AB}\ ^\wedge\ \vec{AD}\right)$ which is $\frac{2\pi}{3}$,

$15\left|\vec{AC}\right|=3\left|\vec{AB}\right|=5\left|\vec{AD}\right|$ then $\cos\left(\vec{BA}\ ^\wedge\ \vec{CD}\right)$ is

(A) $-\frac{\sqrt{14}}{7\sqrt{2}}$ (B) $-\frac{\sqrt{21}}{7\sqrt{3}}$ (C) $\frac{2}{\sqrt{7}}$ (D) $\frac{2\sqrt{7}}{14}$

MVT132

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17. If the two adjacent sides of two rectangles are represented by the vectors $\vec{p}=5\vec{a}-3\vec{b}$; $\vec{q}=-\vec{a}-2\vec{b}$ and $\vec{r}=-4\vec{a}-\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ respectively, then the angle between the vectors $\vec{x}=\frac{1}{3}(\vec{p}+\vec{r}+\vec{s})$ and $\vec{y}=\frac{1}{5}(\vec{r}+\vec{s})$

(A) is $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (B) is $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(C) is $\pi-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (D) cannot be evaluated

MVT133

Paragraph for Question 18 to 20

Consider three vectors $\vec{p}=\hat{i}+\hat{j}+\hat{k}$, $\vec{q}=2\hat{i}+4\hat{j}-\hat{k}$ and $\vec{r}=\hat{i}+\hat{j}+3\hat{k}$ and let $\vec{s}$ be a unit vector, then

18. $\vec{p}$, $\vec{q}$ and $\vec{r}$ are

(A) linearly dependent

(B) can form the sides of a possible triangle

(C) such that the vectors $(\vec{q}-\vec{r})$ is orthogonal to $\vec{p}$

(D) such that each one of these can be expressed as a linear combination of the other two

MVT134

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19. If $(\vec{p}\times\vec{q})\times\vec{r}=u\vec{p}+v\vec{q}+w\vec{r}$, then $(u+v+w)$ equals to

(A) 8 (B) 2 (C) – 2 (D) 4

MVT135

20. the magnitude of the vector $(\vec{p}\cdot\vec{s})(\vec{q}\times\vec{r})+(\vec{q}\cdot\vec{s})(\vec{r}\times\vec{p})+(\vec{r}\cdot\vec{s})(\vec{p}\times\vec{q})$ is

(A) 4 (B) 8 (C) 18 (D) 2

MVT136

Paragraph for Question 21 to 23

The quadratic equation in y, $(a-\sin x-\sin^2 x)y^2+(b-\cos x-\cos^2 x)y+c=0$ is satisfied by each value of λ for which vectors, $\vec{v}_1=\hat{i}+\lambda\hat{j}$, $\vec{v}_2=\lambda\hat{i}+\hat{j}+\hat{k}$ & $\vec{v}_3=\hat{i}+\hat{j}+\lambda\hat{k}$ are linearly dependent. Let $\lambda_0\in Q$ be one of the values of λ.

On the basis of above information, answer the following questions :

21. Maximum value of $a+b+c$ is -

MVT137

22. For $\lambda=\lambda_0$ value of $\left|(\vec{v}_1\times\vec{v}_2)\times\vec{v}_3\right|$ is -

MVT138

23. Number of values of λ for which $\vec{v}_1,\vec{v}_2$ & $\vec{v}_3$ form a triangle –

MVT139

MATCHING LIST TYPE QUESTION

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24.

	List-I		List-II
(I)	P is point in the plane of the triangle ABC. pv's of A, B and C are $\vec{a},\vec{b}$ and $\vec{c}$ respectively with respect to P as the origin. If $(\vec{b}+\vec{c}).(\vec{b}-\vec{c})=0$ and $(\vec{c}+\vec{a}).(\vec{c}-\vec{a})=0$, then w.r.t. the triangle ABC, P is its	(P)	centroid
(II)	If $\vec{a},\vec{b},\vec{c}$ are the position vectors of the three non collinear points A, B and C respectively such that the vector $\vec{V}=\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ is a null vector then w.r.t. the ΔABC, P is its	(Q)	orthocentre
(III)	If P is a point inside the ΔABC such that the vector $\vec{R}=(BC)\overrightarrow{PA}+(CA)\left(\overrightarrow{PB}\right)+(AB)\left(\overrightarrow{PC}\right)$ is a null vector then w.r.t. the ΔABC, P is its	(R)	Incentre
(IV)	If P is a point in the plane of the triangle ABC such that the scalar product $\overrightarrow{PA}.\overrightarrow{CB}$ and $\overrightarrow{PB}.\overrightarrow{AC}$ vanishes, then w.r.t. the ΔABC, P is its	(S)	circumcentre

(A) I $\to$ P; II $\to$ Q; III $\to$ R; IV $\to$ S (B) I $\to$ S; II $\to$ P; III $\to$ R; IV $\to$ Q

(C) I $\to$ R; II $\to$ P; III $\to$ Q; IV $\to$ S (D) I $\to$ R; II $\to$ S; III $\to$ Q; IV $\to$ P

MVT140

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

25.

	Column-I		**Column-II**
(A)	If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar unit vectors, such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ lies in	(P)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(B)	Four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$ Let P_1 and P_2 be planes determined by the pair of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively, then angle between P_1 and P_2 lies in	(Q)	$\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$
		(R)	$\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
(C)	If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $2\vec{a} - \vec{b}$ and $4\vec{a} + 5\vec{b}$ are perpendicular to each other, then angle between $\vec{a}$ and $\vec{b}$ lies in		
(D)	If $\lvert\vec{a}\rvert = 3, \lvert\vec{b}\rvert = 5, \lvert\vec{c}\rvert = 7$ and $\vec{a}+\vec{b}+\vec{c} = \vec{0}$, then angle between $\vec{a}$ and $\vec{b}$ lies in	(S)	$\left(-\frac{\pi}{6}, \frac{7\pi}{12}\right)$

MVT141

Rankers Academy JEE

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EXERCISE (S)

1. A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ radians/sec. If $\vec{\omega}$ points in the direction of $\hat{i}+\hat{j}+\hat{k}$ and the equation to the locus of the points having tangential speed 20 m/sec. is $x^2+y^2+z^2-axy-byz-czx-2=0$, then $(a+b+c)$ is equal to

MVT142

2. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is colliner with $\vec{a}$, then $|\vec{a}+3\vec{b}+6\vec{c}|$ is :

MVT143

3. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a}=\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b}=\frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $\left(2\vec{a}+\vec{b}\right).\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$ is

MVT144

4. If $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$, then $|2\vec{a}+5\vec{b}+5\vec{c}|$ is

MVT145

5. Let $|\vec{a}|=2,|\vec{b}|=1$ and $|\vec{c}|=3$ such that $\vec{a}\wedge\vec{b}=\frac{\pi}{3},\vec{b}\wedge\vec{c}=\frac{\pi}{4}$ and $\vec{c}\wedge\vec{a}=\frac{\pi}{2}$. If $|\vec{a}-2\vec{b}+\vec{c}|$ can be expressed as $\sqrt{p+q\sqrt{2}}$ (where $p,q\in$ I), then $(p+q)$ is

MVT146

Rankers Academy IEE

6. If $\vec{p}=3\vec{a}-5\vec{b}$; $\vec{q}=2\vec{a}+\vec{b}$; $\vec{r}=\vec{a}+4\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ are four vectors such that $\sin\left(\vec{p}\wedge\vec{q}\right)=1$ and $\sin\left(\vec{r}\wedge\vec{s}\right)=1$ then $\sqrt{43}\cos\left(\vec{a}\wedge\vec{b}\right)$ is :

MVT147

7. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c}=\hat{a}+2\hat{b}$ and $\vec{d}=5\hat{a}-4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is :

MVT148

8. Two adjacent sides of a parallelogram $ABCD$ are given by $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ and $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB, then the cosine of the angle α is given by -

MVT149

9. Let $\vec{r}=\sin x\left(\vec{a}\times\vec{b}\right)+\cos y\left(\vec{b}\times\vec{c}\right)+2\left(\vec{c}\times\vec{a}\right)$, where $\vec{a},\vec{b}$ & $\vec{c}$ are non-zero and non-coplanar vectors. If $\vec{r}$ is orthogonal to $\vec{a}+\vec{b}+\vec{c}$, then minimum value of $\frac{4}{\pi^2}\left(x^2+y^2\right)$ is

MVT150

10. If three points $\left(2\vec{p}-\vec{q}+3\vec{r}\right),\left(\vec{p}-2\vec{q}+\alpha\vec{r}\right)$ and $\left(\beta\vec{p}-5\vec{q}\right)$ (where $\vec{p},\vec{q},\vec{r}$ are non-coplanar vectors) are collinear, then the value of $\frac{1}{\alpha+\beta}$ is

MVT151

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let $\vec{\alpha}=(\lambda-2)\vec{a}+\vec{b}$ and $\vec{\beta}=(4\lambda-2)\vec{a}+3\vec{b}$ be two given vectors where vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is: **[JEE (Main) 2019]**

(1) –3 (2) 4 (3) 3 (4) –4

MVT026

2. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and $\vec{a}\times(\vec{b}\times\vec{c})=\frac{1}{2}\vec{b}$, then $|\alpha-\beta|$ is equal to : **[JEE (Main) 2019]**

(1) 60° (2) 30° (3) 90° (4) 45°

MVT027

3. Let $\vec{a}=3\hat{i}+2\hat{j}+x\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$, for some real x. Then $|\vec{a}\times\vec{b}|=r$ is possible if : **[JEE (Main) 2019]**

(1) $3\sqrt{\frac{3}{2}}<r<5\sqrt{\frac{3}{2}}$ (2) $0<r\le\sqrt{\frac{3}{2}}$ (3) $\sqrt{\frac{3}{2}}<r\le3\sqrt{\frac{3}{2}}$ (4) $r\ge5\sqrt{\frac{3}{2}}$

MVT028

4. Let $\vec{\alpha}=3\hat{i}+\hat{j}$ and $\vec{\beta}=2\hat{i}-\hat{j}+3\hat{k}$. If $\vec{\beta}=\vec{\beta}_1-\vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1\times\vec{\beta}_2$ is equal to **[JEE (Main) 2019]**

Rankers Academy JEE

(1) $-3\hat{i}+9\hat{j}+5\hat{k}$ (2) $3\hat{i}-9\hat{j}-5\hat{k}$ (3) $\frac{1}{2}(-3\hat{i}+9\hat{j}+5\hat{k})$ (4) $\frac{1}{2}(3\hat{i}-9\hat{j}+5\hat{k})$

MVT029

5. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three units vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If $\lambda=\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ and $\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$, then the ordered pair, $(\lambda,\vec{d})$ is equal to: **[JEE (Main) 2020]**

(1) $\left(-\frac{3}{2},3(\vec{a}\times\vec{b})\right)$ (2) $\left(-\frac{3}{2},3(\vec{c}\times\vec{b})\right)$ (3) $\left(\frac{3}{2},3(\vec{b}\times\vec{c})\right)$ (4) $\left(\frac{3}{2},3(\vec{a}\times\vec{c})\right)$

MVT030

6. Let the volume of a parallelepiped whose coterminous edges are given by $\vec{u}=\hat{i}+\hat{j}+\lambda\hat{k},\vec{v}=\hat{i}+\hat{j}+3\hat{k}$ and $\vec{w}=2\hat{i}+\hat{j}+\hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos\theta$ can be **[JEE (Main) 2020]**

(1) $\frac{7}{6\sqrt{3}}$ (2) $\frac{5}{7}$ (3) $\frac{7}{6\sqrt{6}}$ (4) $\frac{5}{3\sqrt{3}}$

MVT031

7. Let x_0 be the point of local maxima of $f(x)=\vec{a}.(\vec{b}\times\vec{c})$, where $\vec{a}=x\hat{i}-2\hat{j}+3\hat{k}$, $\vec{b}=-2\hat{i}+x\hat{j}-\hat{k}$ and $\vec{c}=7\hat{i}-2\hat{j}+x\hat{k}$. Then the value of $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ at $x=x_0$ is : **[JEE (Main) 2020]**

(1) –30 (2) 14 (3) –4 (4) –22

MVT032

For More Join: @IITJEE_Advanced

8. If the volume of a parallelepiped, whose coterminous edges are given by the vectors $\vec{a}=\hat{i}+\hat{j}+n\hat{k}$, $\vec{b}=2\hat{i}+4\hat{j}-n\hat{k}$ and $\vec{c}=\hat{i}+n\hat{j}+3\hat{k}$ ($n \geq 0$), is 158 cu. units, then : **[JEE (Main) 2020]**

(1) $\vec{a}\cdot\vec{c}=17$ (2) $\vec{b}\cdot\vec{c}=10$ (3) $n=7$ (4) $n=9$

MVT033

9. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three-unit vectors such that $|\vec{a}-\vec{b}|^2+|\vec{a}-\vec{c}|^2=8$. Then $|\vec{a}+2\vec{b}|^2+|\vec{a}+2\vec{c}|^2$ is equal to. **[JEE (Main) 2020]**

MVT034

10. Let the position vectors of points 'A' and 'B' be $\hat{i}+\hat{j}+\hat{k}$ and $2\hat{i}+\hat{j}+3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio λ : 1 (λ > 0). If O is the origin and $\overrightarrow{OB}\cdot\overrightarrow{OP}-3|\overrightarrow{OA}\times\overrightarrow{OP}|^2=6$, then λ is equal to. **[JEE (Main) 2020]**

MVT035

11. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}|=\sqrt{3}$, $|\vec{b}|=5$, $\vec{b}\cdot\vec{c}=10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b}\times\vec{c}$, then $|\vec{a}\times(\vec{b}\times\vec{c})|$ is equal to. **[JEE (Main) 2020]**

MVT036

12. If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|$ is _. **[JEE (Main) 2020]**

MVT037

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13. If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x}+\vec{y}|=|\vec{x}|$ and $2\vec{x}+\lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is _____. **[JEE (Main) 2020]**

MVT038

14. Let $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i}+2\hat{j}-\hat{k}$ and its projection on $\vec{a}$ is 19 units, then $|2\vec{v}|^2$ is equal to. **[JEE (Main) 2021]**

MVT039

15. Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=2\hat{i}-3\hat{j}+5\hat{k}$. If $\vec{r}\times\vec{a}=\vec{b}\times\vec{r}$, $\vec{r}.\left(\alpha\hat{i}+2\hat{j}+\hat{k}\right)=3$ and $\vec{r}.\left(2\hat{i}+5\hat{j}-\alpha\hat{k}\right)=-1$, $\alpha \in R$, then the value of $\alpha+|\vec{r}|^2$ is equal to : **[JEE (Main) 2021]**

(1) 9 (2) 15 (3) 13 (4) 11

MVT040

16. Let a vector $\alpha\hat{i}+\beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i}+\hat{j}$ by an angle 45° about the origin in counter clockwise direction in the first quadrant. Then the area of triangle having vertices (α, β), (0, β) and (0, 0) is equal to **[JEE (Main) 2021]**

(1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{\sqrt{2}}$ (4) $2\sqrt{2}$

MVT041

For More Join: @IITJEE_Advanced

17. Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. If the vector $\vec{x}$ is perpendicular to $\left(3\hat{i}+2\hat{j}-\hat{k}\right)$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to.

[JEE (Main) 2021]

MVT042

18. A vector $\vec{a}$ has components $3p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p+1$ and $\sqrt{10}$, then a value of p is equal to:

[JEE (Main) 2021]

(1) 1 (2) $-\frac{5}{4}$ (3) $\frac{4}{5}$ (4) –1

MVT043

19. For $p>0$, a vector $\vec{v}_2=2\hat{i}+(p+1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1=\sqrt{3}p\hat{i}+\hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta=\frac{(\alpha\sqrt{3}-2)}{(4\sqrt{3}+3)}$, then the value of α is equal to.

[JEE (Main) 2021]

MVT044

20. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a}+\vec{b}+\vec{c}$. Then $36\cos^2 2\theta$ is equal to _____. **[JEE (Main) 2021]**

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MVT045

21. Let three vectors $\vec{a},\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c},\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$. Then which one of the following is **not** true ? **[JEE (Main) 2021]**

(1) $\vec{a}\times((\vec{b}+\vec{c})\times(\vec{b}-\vec{c}))=\vec{0}$ (2) Projection of $\vec{a}$ on $(\vec{b}\times\vec{c})$ is 2

(3) $[\vec{a}\ \ \vec{b}\ \ \vec{c}]+[\vec{c}\ \ \vec{a}\ \ \vec{b}]=8$ (4) $|3\vec{a}+\vec{b}-2\vec{c}|^2=51$

MVT046

22. Let $\vec{b}=\hat{i}+\hat{j}+\lambda\hat{k},\lambda\in R$. If $\vec{a}$ is a vector such that $\vec{a}\times\vec{b}=13\hat{i}-\hat{j}-4\hat{k}$ and $\vec{a}\cdot\vec{b}+21=0$, then $(\vec{b}-\vec{a})\cdot(\hat{k}-\hat{j})+(\vec{b}+\vec{a})\cdot(\hat{i}-\hat{k})$ is equal to **[JEE (Main) 2022]**

MVT047

23. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $a_i>0, i=1,2,3$ be a vector which makes equal angles with the coordinates axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i}+4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a},\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i}+4\hat{j}$ is equal to **[JEE (Main) 2022]**

(1) $\sqrt{7}$ (2) $\sqrt{2}$ (3) 2 (4) 7

MVT048

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24. Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\hat{j}=7$, then $\vec{v}.\left(\hat{i}+\hat{k}\right)$ is equal to : **[JEE (Main) 2022]**

(1) 6 (2) 7 (3) 8 (4) 9

MVT049

25. If $\vec{a}\cdot\vec{b}=1$, $\vec{b}\cdot\vec{c}=2$ and $\vec{c}\cdot\vec{a}=3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right),\vec{b}\times\left(\vec{c}\times\vec{a}\right),\vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

[JEE (Main) 2022]

(1) 0 (2) $-6\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)$ (3) $12\vec{c}\cdot\left(\vec{a}\times\vec{b}\right)$ (4) $-12\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$

MVT050

26. Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonal of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute. $|\vec{a}|=1$ and $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$. If $\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is : **[JEE (Main) 2022]**

(1) $\frac{\pi}{4}$ (2) $-\frac{\pi}{4}$ (3) $\frac{5\pi}{6}$ (4) $\frac{3\pi}{4}$

MVT051

27. Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$. Then $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ is equal to **[JEE (Main) 2023]**

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(1) 4 (2) 3 (3) 2 (4) $\frac{5}{2}$

MVT152

28. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+\mu\hat{k}$ and $3\hat{i}-4\hat{j}+5\hat{k}$, where $\lambda-\mu=5$, are coplanar, then $\sum_{(\lambda,\mu)\in S}80(\lambda^2+\mu^2)$ is equal to : **[JEE (Main) 2023]**

(1) 2370 (2) 2130 (3) 2290 (4) 2210

MVT153

29. An arc PQ of a circle subtends a right angle at its centre O. The mid point of the arc PQ is R. If $\overrightarrow{OP}=\vec{u}$, $\overrightarrow{OR}=\vec{v}$ and $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$, then α, β^2 are the roots of the equation **[JEE (Main) 2023]**

(1) $x^2-x-2=0$ (2) $3x^2+2x-1=0$

(3) $x^2+x-2=0$ (4) $3x^2-2x-1=0$

MVT154

30. For any vector $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, with $10|a_i|<1, i=1,2,3$, consider the following statements:

(A) : $\max\{|a_1|,|a_2|,|a_3|\}\le|\vec{a}|$

(B) : $|\vec{a}|\le 3\max\{|a_1|,|a_2|,|a_3|\}$ **[JEE (Main) 2023]**

(1) Only (B) is true (2) Only (A) is true

(3) Neither (A) nor (B) is true (4) Both (A) and (B) are true

MVT155

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31. Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$, $\vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a}\times\left(\vec{b}+\vec{c}\right)+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$. If $\vec{a}\cdot\vec{c}=13$, then $\left(24-\vec{b}\cdot\vec{c}\right)$ is equal to _____. **[JEE (Main) 2024]**

MVT255

32. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}|\geq 6, \vec{a}\cdot\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to: **[JEE (Main) 2024]**

(1) $\frac{9}{2}\left(6-\sqrt{6}\right)$ (2) $\frac{3}{2}\sqrt{3}$ (3) $\frac{3}{2}\sqrt{6}$ (4) $\frac{9}{2}\left(6+\sqrt{6}\right)$

MVT256

33. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{b}=\left(\left(\vec{a}\times\left(\hat{i}+\hat{j}\right)\right)\times\hat{i}\right)\times i$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is : **[JEE (Main) 2024]**

(1) $\frac{1}{5}$ (2) 2 (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

MVT257

34. Consider three vectors $\vec{a},\vec{b},\vec{c}$. Let $|\vec{a}|=2, |\vec{b}|=3$ and $\vec{a}=\vec{b}\times\vec{c}$. If $a\in\left[0,\frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c}-\vec{a}|^2$ is equal to : **[JEE (Main) 2024]**

(1) 110 (2) 105 (3) 124 (4) 121

MVT258

Rankers Academy JEE

35. Let $\vec{a}=2\hat{i}+5\hat{j}-\hat{k}$, $\vec{b}=2\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{c}$ be three vectors such that $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\hat{b}+\hat{i}\right)=\vec{a}\times\left(\vec{c}+\hat{i}\right)$. $\vec{a}.\vec{c}=-29$, then $\vec{c}\cdot\left(-2\hat{i}+\hat{j}+\hat{k}\right)$ is equal to : **[JEE (Main) 2024]**

(1) 10 (2) 5 (3) 15 (4) 12

MVT259

36. Let $\vec{a}=\hat{i}-3\hat{j}+7\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$ and $\vec{c}$ be a vector such that $\left(\vec{a}+2\vec{b}\right)\times\vec{c}=3\left(\vec{c}\times\vec{a}\right)$. If $\vec{a}\cdot\vec{c}=130$, then $\vec{b}\cdot\vec{c}$ is equal to _____. **[JEE (Main) 2024]**

MVT260

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Let $\overrightarrow{PR}=3\hat{i}+\hat{j}-2\hat{k}$ and $\overrightarrow{SQ}=\hat{i}-3\hat{j}-4\hat{k}$ determine diagonals of a parallelogram $PQRS$ and $\overrightarrow{PT}=\hat{i}+2\hat{j}+3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}$, $\overrightarrow{PQ}$ and $\overrightarrow{PS}$ is **[JEE (Advanced) 2013]**

(A) 5 (B) 20 (C) 10 (D) 30

MVT074

2. Perpendicular are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane $x+y+z=3$. The feet of perpendiculars lie on the line **[JEE (Advanced) 2013]**

(A) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$ (B) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$ (C) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$ (D) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

MVT075

3. A line l passing through the origin is perpendicular to the lines

$l_1:(3+t)\,\hat{i}+(-1+2t)\,\hat{j}+(4+2t)\,\hat{k}, -\infty<t<\infty$

$l_2:(3+2s)\,\hat{i}+(3+2s)\,\hat{j}+(2+s)\,\hat{k}, -\infty<s<\infty$

Then, the coordinate(s) of the point(s) on l_2 at a distance of $\sqrt{17}$ from the point of intersection of l and l_1 is(are) **[JEE (Advanced) 2013]**

(A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ (B) $(-1,-1,0)$ (C) $(1,1,1)$ (D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

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MVT076

4. Consider the set of eight vectors $V=\left\{a\hat{i}+b\hat{j}+c\hat{k}\ :\ a,b,c\in\{-1,1\}\right\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is **[JEE (Advanced) 2013]**

MVT077

5. Match **List I** with **List II** and select the correct answer using the code given below the lists :

List - I	**List - II**
(P) Volume of parallelepiped determined by vectors $\vec{a},\vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2(\vec{a}\times\vec{b}),3(\vec{b}\times\vec{c})$ and $(\vec{c}\times\vec{a})$ is	(1) 100
(Q) Volume of parallelepiped determined by vectors $\vec{a},\vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3(\vec{a}+\vec{b}),(\vec{b}+\vec{c})$ and $2(\vec{c}+\vec{a})$ is	(2) 30
(R) Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a}+3\vec{b})$ and $(\vec{a}-\vec{b})$ is	(3) 24
(S) Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a}+\vec{b})$ and $\vec{a}$ is	(4) 60

For More Join: @IITJEE_Advanced

Codes : [JEE (Advanced) 2013]

	P	Q	R	S
(A)	4	2	3	1
(B)	2	3	1	4
(C)	3	4	1	2
(D)	1	4	3	2

MVT078

6. Let $\vec{x},\vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y}\times\vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z}\times\vec{x}$, then [JEE (Advanced) 2014]

(A) $\vec{b}=(\vec{b}.\vec{z})(\vec{z}-\vec{x})$ (B) $\vec{a}=(\vec{a}.\vec{y})(\vec{y}-\vec{z})$ (C) $\vec{a}.\vec{b}=-(\vec{a}.\vec{y})(\vec{b}.\vec{z})$ (D) $\vec{a}=(\vec{a}.\vec{y})(\vec{z}-\vec{y})$

MVT079

7. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$. If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p,q and r are scalars, then the value of $\frac{p^2+2q^2+r^2}{q^2}$ is

[JEE (Advanced) 2014]

MVT080

8.

List I	List II
(P) Let $y(x)=\cos\left(3\cos^{-1}x\right)$, $x\in[-1,1]$, $x\neq\pm\frac{\sqrt{3}}{2}$. Then $\frac{1}{y(x)}\left\{\left(x^2-1\right)\frac{d^2y(x)}{dx^2}+x\frac{dy(x)}{dx}\right\}$ equals	(1) 1
(Q) Let $A_1,A_2,.....,A_n$ $(n>2)$ be the vertices of a regular polygon of n sides with its centre at the origin. Let $\vec{a}_k$ be the position vector of the point $A_k, k=1,2,....,n$. If $\left\|\sum_{k=1}^{n-1}(\vec{a_k}\times\vec{a_{k+1}})\right\|=\left\|\sum_{k=1}^{n-1}(\vec{a_k}.\vec{a_{k+1}})\right\|$, then the minimum value of n is	(2) 2
(R) If the normal from the point $P(h,1)$ on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ is perpendicular to the line $x+y=8$, then the value of h is	(3) 8
(S) Number of positive solutions satisfying the equation is $\tan^{-1}\left(\frac{1}{2x+1}\right)+\tan^{-1}\left(\frac{1}{4x+1}\right)=\tan^{-1}\left(\frac{2}{x^2}\right)$	(4) 9

Code: [JEE (Advanced) 2014]

	P	Q	R	S
(A)	4	3	2	1
(B)	2	4	3	1
(C)	4	3	1	2
(D)	2	4	1	3

MVT081

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9. Let ΔPQR be a triangle. Let $\vec{a}=\overrightarrow{QR}, \vec{b}=\overrightarrow{RP}$ and $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|=12$, $|\vec{b}|=4\sqrt{3}$ and $\vec{b}.\vec{c}=24$, then which of the following is(are) true? **[JEE (Advanced) 2015]**

(A) $\frac{|\vec{c}|^2}{2}-|\vec{a}|=12$

(B) $\frac{|\vec{c}|^2}{2}+|\vec{a}|=30$

(C) $|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}|=48\sqrt{3}$

(D) $\vec{a}.\vec{b}=-72$

MVT082

10.

Column- I	**Column-II**
(A) In a triangle ΔXYZ, let a,b and c be the lengths of the sides opposite to the angles X,Y and Z, respectively. If $2\left(a^2-b^2\right)=c^2$ and $\lambda=\frac{\sin(X-Y)}{\sin Z}$, then possible values of n for which $\cos\left(n\pi\lambda\right)=0$ is (are)	(P) 1
(B) In a triangle ΔXYZ, let a,b and c be the lengths of the sides opposite to the angles X,Y and Z, respectively. If $1+\cos 2X-2\cos 2Y=2\sin X\sin Y$, then possible value(s) of $\frac{a}{b}$ is (are)	(Q) 2
(C) In R^2, let $\sqrt{3}\hat{i}+\hat{j}, \hat{i}+\sqrt{3}\hat{j}$ and $\beta\hat{i}+(1-\beta)\hat{j}$ be the position vectors of X,Y and Z with respect to the origin O, respectively. If the distance of Z from the bisector of the acute angle of $\overrightarrow{OX}$ and $\overrightarrow{OY}$ is, $\frac{3}{\sqrt{2}}$ then possible value(s) of $\|\beta\|$ is (are)	(R) 3
(D) Suppose that $F(\alpha)$ denotes the area of the region bounded by $x=0$, $x=2$, $y^2=4x$ and $y=\|\alpha x-1\|+\|\alpha x-2\|+\alpha x$, where $\alpha\in\{0,1\}$. Then the value(s) of $F(\alpha)+\frac{8}{3}\sqrt{2}$, when $\alpha = 0$ and $\alpha = 1$, is (are)	(S) 5

[JEE (Advanced) 2015]

MVT083

11. Suppose that $\vec{p},\vec{q}$ and $\vec{r}$ are three non-coplanar vectors in R^3. Let the components of a vector $\vec{s}$ along $\vec{p},\vec{q}$ and $\vec{r}$ be 4, 3 and 5, respectively. If the components of this vector $\vec{s}$ along $(-\vec{p}+\vec{q}+\vec{r}),(\vec{p}-\vec{q}+\vec{r})$ and $(-\vec{p}-\vec{q}+\vec{r})$ are x, y and z, respectively, then the value of $2x+y+z$ is **[JEE (Advanced) 2015]**

MVT084

For More Join: @IITJEE_Advanced

12. Let $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ be a unit vector in $\mathbb{R}^2$ and $\hat{w}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$. Given that there exists a vector $\vec{v}$ in $\mathbb{R}^3$ such that $|\hat{u}\times\vec{v}|=1$ and $\hat{w}.(\hat{u}\times\vec{v})=1$. Which of the following statement(s) is(are) correct?

[JEE(Advanced)-2016]

(A) There is exactly one choice for such $\vec{v}$ (B) There are infinitely many choice for such $\vec{v}$

(C) If $\hat{u}$ lies in the xy-plane then $|u_1|=|u_2|$ (D) If $\hat{u}$ lies in the xz-plane then $2|u_1|=|u_3|$

MVT085

13. Let O be the origin and let PQR be an arbitrary triangle. The point S is such that $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$. Then the triangle PQR has S as its

[JEE(Advanced) 2017]

(A) incentre (B) orthocenter (C) circumcenter (D) centroid

MVT086

Paragraph : (Q.14 to 15)

Let O be the origin, and $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ be three-unit vectors in the directions of the sides $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$, respectively, of a triangle PQR.

14. $\left|\overrightarrow{OX}\times\overrightarrow{OY}\right|=$ **[JEE (Advanced) 2017]**

(A) $\sin(Q+R)$ (B) $\sin(P+R)$ (C) $\sin 2R$ (D) $\sin(P+Q)$

MVT087

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15. If the triangle PQR varies, then the minimum value of $\cos(P+Q)+\cos(Q+R)+\cos(R+P)$ is

[JEE (Advanced) 2017]

(A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $-\frac{5}{3}$

MVT088

16. Let $\vec{a}$ and $\vec{b}$ be two-unit vectors such that $\vec{a}.\vec{b}=0$. For some $x,y\in\mathbb{R}$, let $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$. If $|\vec{c}|=2$ and the vector $|\vec{c}|$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2\alpha$ is.

[JEE (Advanced) 2018]

MVT089

17. Three lines

$L_1:\vec{r}=\lambda\hat{i},\ \lambda\in\mathbb{R}$,

$L_2:\vec{r}=\hat{k}+\mu\hat{j},\ \mu\in\mathbb{R}$ and

$L_3:\vec{r}=\hat{i}+\hat{j}+\nu\hat{k},\ \nu\in\mathbb{R}$

are given. For which point(s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P,Q and R are collinear **[JEE (Advanced) 2019]**

(A) $\hat{k}+\hat{j}$ (B) $\hat{k}$ (C) $\hat{k}+\frac{1}{2}\hat{j}$ (D) $\hat{k}-\frac{1}{2}\hat{j}$

MVT090

For More Join: @IITJEE_Advanced

18. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ be two vectors. Consider a vector $\vec{c}=\alpha\vec{a}+\beta\vec{b}$, $\alpha,\beta \in \mathbb{R}$. If the projection of $\vec{c}$ on the vector $\left(\vec{a}+\vec{b}\right)$ is $3\sqrt{2}$, then the minimum value of $\left(\vec{c}-\left(\vec{a}\times\vec{b}\right)\right).\vec{c}$ equals

[JEE (Advanced) 2019]

MVT091

19. In a triangle PQR, let $\vec{a}=\overrightarrow{QR}, \vec{b}=\overrightarrow{RP}$ *and* $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|$ = 3, $|\vec{b}|$ = 4 and $\frac{\vec{a}.(\vec{c}-\vec{b})}{\vec{c}.(\vec{a}-\vec{b})}=\frac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$, then the value of $|\vec{a}\times\vec{b}|^2$ is.

[JEE (Advanced) 2020]

MVT092

20. Let a and b be positive real numbers. Suppose $\overrightarrow{PQ}=a\hat{i}+b\hat{j}$ and $\overrightarrow{PS}=a\hat{i}-b\hat{j}$ are adjacent sides of a parallelogram $PQRS$. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w}=\hat{i}+\hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If $|\vec{u}|+|\vec{v}|=|\vec{w}|$ and if the area of the parallelogram $PQRS$ is 8, then which of the following statements is/are TRUE ?

[JEE (Advanced) 2020]

(A) $a+b=4$

(B) $a-b=2$

(C) The length of the diagonal PR of the parallelogram $PQRS$ is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$

MVT093

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21. Let $\vec{u},\vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u}.\vec{w}=1$, $\vec{v}.\vec{w}=1$, $\vec{w}.\vec{w}=4$. If the volume of the parallelepiped, whose adjacent sides are represented by the vectors $\vec{u},\vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u}+5\vec{v}|$ is.

[JEE (Advanced) 2021]

MVT094

22. Let O be the origin and $\overrightarrow{OA}=2\hat{i}+2\hat{j}+\hat{k}$, $\overrightarrow{OB}=\hat{i}-2\hat{j}+2\hat{k}$ and $\overrightarrow{OC}=\frac{1}{2}(\overrightarrow{OB}-\lambda\overrightarrow{OA})$ for some $\lambda > 0$. If $|\overrightarrow{OB}\times\overrightarrow{OC}|=\frac{9}{2}$, then which of the following statements is (are) **TRUE**?

[JEE (Advanced) 2021]

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\frac{3}{2}$

(B) Area of the triangle OAB is $\frac{9}{2}$

(C) Area of the triangle ABC is $\frac{9}{2}$

(D) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\frac{\pi}{3}$

MVT095

For More Join: @IITJEE_Advanced

23. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes. Let

$\vec{a} = 3\hat{i} + \hat{j} - \hat{k},$

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}, \quad b_2, b_3 \in \mathbb{R},$

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}, \quad c_1, c_2, c_3 \in \mathbb{R}$

be three vectors such that $b_2 b_3 > 0$, $\vec{a} \cdot \vec{b} = 0$ and

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}.$$

Then, which of the following is/are TRUE ? **[JEE (Advanced) 2022]**

(A) $\vec{a} \cdot \vec{c} = 0$ (B) $\vec{b} \cdot \vec{c} = 0$ (C) $|\vec{b}| > \sqrt{10}$ (D) $|\vec{c}| \leq \sqrt{11}$

MVT096

24. Let P be the plane $\sqrt{3}x + 2y + 3z = 16$ and let

$$S = \left\{ \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k} : \alpha^2 + \beta^2 + \gamma^2 = 1 \text{ and the distance of } (\alpha, \beta, \gamma) \text{ from the plane } P \text{ is } \frac{7}{2} \right\}.$$

Let $\vec{u}, \vec{v}$ and $\vec{w}$ be three distinct vectors in S such that $|\vec{u} - \vec{v}| = |\vec{v} - \vec{w}| = |\vec{w} - \vec{u}|$. Let V be the volume of the parallelepiped determined by vectors $\vec{u}, \vec{v}$ and $\vec{w}$. Then the value of $\frac{80}{\sqrt{3}}V$ is

[JEE (Advanced) 2023]

MVT156

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25. Let ℓ_1 and ℓ_2 be the lines $\vec{r}_1 = \lambda(\hat{i} + \hat{j} + \hat{k})$ and $\vec{r}_2 = (\hat{j} - \hat{k}) + \mu(\hat{i} + \hat{k})$, respectively. Let X be the set of all the planes H that contain the line ℓ_1. For a plane H, let $d(H)$ denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be plane in X for which $d(H_0)$ is the maximum value of $d(H)$ as H varies over all planes in X.

Match each entry in **List-I** to the correct entries in **List-II.**

List-I	**List-II**
(P) The value of $d(H_0)$ is	(1) $\sqrt{3}$
(Q) The distance of the point (0,1,2) from H_0 is	(2) $\frac{1}{\sqrt{3}}$
(R) The distance of origin from H_0 is	(3) 0
(S) The distance of origin from the point of intersection of planes $y = z$, $x = 1$ and H_0 is	(4) $\sqrt{2}$
	(5) $\frac{1}{\sqrt{2}}$

The correct option is : **[JEE (Advanced) 2023]**

(A) (P) → (2) (Q) → (4) (R) → (5) (S) → (1) (B) (P) → (5) (Q) → (4) (R) → (3) (S) → (1)

(C) (P) → (2) (Q) → (1) (R) → (3) (S) → (2) (D) (P) →(5) (Q) → (1) (R) → (4) (S) → (2)

MVT157

For More Join: @IITJEE_Advanced

26. Let the position vectors of the points P, Q, R and S be $\vec{a}=\hat{i}+2\hat{j}-5\hat{k}$, $\vec{b}=3\hat{i}+6\hat{j}+3\hat{k}$, $\vec{c}=\frac{17}{5}\hat{i}+\frac{16}{5}\hat{j}+7\hat{k}$ and $\vec{d}=2\hat{i}+\hat{j}+\hat{k}$, respectively. Then which of the following statements is true?

[JEE (Advanced) 2023]

(A) The points P, Q, R and S are **NOT** coplanar

(B) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR internally in the ratio 5 : 4

(C) $\frac{\vec{b}+2\vec{d}}{3}$ is the position vector of a point which divides PR externally in the ratio 5 : 4

(D) The square of the magnitude of the vector $\vec{b}\times\vec{d}$ is 95

MVT158

27. Let $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ and $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$ be three vectors, where $\alpha,\beta\in\mathbb{R}-\{0\}$ and O denotes the origin. If $\left(\overrightarrow{OP}\times\overrightarrow{OQ}\right).\overrightarrow{OR}=0$ and the point $(\alpha, \beta, 2)$ lies on the plane $3x + 3y - z + l = 0$, then the value of l is .....

[JEE (Advanced) 2024]

MVT253

28. Let $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{q}=\hat{i}-\hat{j}+\hat{k}$. If for some real numbers α, β and γ, we have $15\hat{i}+10\hat{j}+6\hat{k}=\alpha(2\vec{p}+\vec{q})+\beta(\vec{p}-2\vec{q})+\gamma(\vec{p}\times\vec{q})$, then the value of γ is _____.

[JEE (Advanced) 2024]

MVT254

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	C	A	C	D	A	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	C	D	D	B	A	C	D	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C	A,C,D	C,D	A,B,C,D	A,C	B,C,D	A,B,C	B,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,C,D	A,B	A,C	B,D	A,B,D	C	B	C	B	A
Que.	21	22	23	24	25					
Ans.	2.41 or 2.42	2.44 or 2.45	0.00	B	(A) → (R); (B) → (Q,S); (C) → (Q,R); (D) → (P,Q,R,S)					

EXERCISE (S)

1.	3	**2.**	0	**3.**	5	**4.**	3	**5.**	7
6.	3.80	**7.**	1.04 or 1.05	**8.**	0.45 or 0.46	**9.**	5.00	**10.**	4.00

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	4	3	1	1	4	2	2	0.8
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	30	4	1	1494	2	1	486	4	6	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	14	2	4	1	4	2	3	1	4
Que.	31	32	33	34	35	36				
Ans.	46	4	2	3	2	30				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	D	B,D	32	C	A,B,C	4	A	A,C,D	
Que.	10					11	12	13	14	15
Ans.	A → P,R,S; B → P; C → P,Q ; D → S, T					Bonus	B,C	B	D	B
Que.	16	17	18	19	20	21	22	23	24	25
Ans.	3	C,D	18	108	A,C	7	A,B,C	B,C,D	45	B
Que.	26	27	28							
Ans.	B	5	2							

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